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The Editor-in-Chief
Chemico-Biological Interactions (Elsevier)

Dear Editor-in-Chief,

On behalf of all authors, I am pleased to submit our original research article, “Network-based prioritization of venom–host interactions potentially relevant to Russell’s viper–associated thrombotic microangiopathy” by Xiuzhen Pan, Xianggui Zeng, Jiali Li, and Wenhui Chen, for consideration in Chemico-Biological Interactions.

Russell’s viper envenoming can trigger thrombotic microangiopathy (TMA), yet the molecular chain from venom toxins to the TMA phenotype remains fragmented. We built a documented-analysis-plan–guided, multi-layer computational pipeline linking 63 curated *Daboia russelii* / *D. siamensis* toxins to host biology: 447 evidence-tiered toxin–target edges (15 literature-curated direct targets), six-source disease-gene integration (1,042 core genes), network topology, dual-platform enrichment, protein–protein docking on three platforms with orthogonal NMA/PRODIGY descriptors, small-molecule docking of the SVMP inhibitors batimastat/marimastat and the PLA2 inhibitor varespladib, cross-species transcriptome re-analysis, network robustness testing, and AOP-framework integration.

Basic findings: the coagulation/fibrinolysis axis dominates every evidence layer (direct targets, hub topology, enrichment — KEGG:04610, $P = 3.9 \times 10^{-18}$, with all eleven driver genes on the coagulation arm); endothelial/inflammatory activation forms a second, transcriptome-supported pillar; and complement involvement, a widely assumed mechanism, is not supported at the direct-target or enrichment layers and is reported honestly as a negative/secondary finding. Structural analyses converged on a factor-X-activation interface for RVV-X and a template-grade svVEGF–VEGFR2 interaction, and the hydroxamate inhibitors chelated the RVV-X catalytic zinc with correct geometry.

We believe this work fits the journal’s scope: it is mechanistic toxicology of chemically characterized venom toxins, addresses toxin–protein and inhibitor–protein interactions at the molecular level, and is directly relevant to a WHO-listed neglected tropical disease. The entire pipeline is open and reproducible: code and result tables are archived on GitHub and Zenodo (DOIs in the manuscript), with a version-by-version change log and a formal Supplementary Methods and Tables S1–S10 package.

The manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have read and approved the submission and declare no competing interests. No new human or animal experiments were conducted; the study reanalyzes publicly available data. Research data are deposited and linked in accordance with the journal’s research-data policy. The use of AI-assisted tools in the writing process is declared in the manuscript. We would be glad to suggest qualified reviewers through the submission system.

Thank you for considering our work. We look forward to your response.

Sincerely,

Wenhui Chen

on behalf of all authors